Date: 02/03/2024

To: Dr. Terry Xu

From: Mikey Wiesecke MEGR 3152 Section L09

Subject: Analysis of Impact Test of Steel and Aluminum

English Grade Cover Sheet

GRADING MATRIX (ORIGINAL)

Overall Precision and Clarity of Content 20%	0/**5**
Sophistication of Language Mechanics 20%	0/**5**
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Equations, Tables, and Figures 20%	0/**5**
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Was the revision submitted by the deadline? (If no, revision not graded)	Yes
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If you need extra help with the revision process, please feel free to schedule an appointment with Justin Cary. You can reach him at jcary1@uncc.edu any time to schedule your online appointment.

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Abstract

The objective of this experiment was to assess the mechanical properties of both 1018-Cold Finished Steel and 6061-T6 Aluminum in response to temperature. The impact toughness was specifically evaluated by using a Riehle PI-2 Impact Testing Machine. Samples of both metals were cooled and heated from a range of -200 to 200 degrees Celsius. The results came out to show that the steel was extremely sensitive to temperature while the aluminum did not display any differences from the temperature. However, this did not mean that the aluminum performed better but rather it was more consistent and thermally stable. The steel outperformed the aluminum in temperatures above zero degrees Celsius, but turned brittle below this temperature. This is called a brittle to ductile transition temperature which steel exhibits. This was evident in the data as well as the physical steel samples which only partially broke beyond this transition temperature. The difference in results was boiled down to the crystalline structure of each metal. Aluminum, which showed no reaction to temperature, has an FCC structure versus the BCC structure of 1018 steel. As a rule, FCC structures are more thermally stable and BCC structures are less thermally stable. That said, steel still had the greatest energy absorption out the two metals and consequently the greater toughness. In addition to this, steel can also be made to be temperature resistant, as in stainless steel. In conclusion, aluminum has a unique resistance to temperature that is not seen in regular steel. While incredibly tough, steel is also extremely sensitive to temperature and is capable of going from brittle to ductile.

Objective

The purpose of this experiment is to observe the effects of temperature on the toughness of 1018-Cold Finished Steel and 6061-T6 Aluminum. The effects of temperature can be adverse and detrimental to the structure of a component.

Introduction

Metal such as steel and aluminum have different sensitivity to temperature. In general, colder temperatures will make a material more brittle and higher temperature will make a material more ductile. The test performed in this experiment aims to quantify this effect of temperature with the use of a Riehle PI-2 impact testing machine. The process involves measuring the energy lost from breaking or impacting a sample. Plotting this data against temperature can visualize the effects of the temperature and be used to compare varied materials sensitivities. Engineers can make better decisions with material selection in different applications aside from simple stress strain. This experiment will operate at temperatures ranging from -200 degrees to 200 degrees Celsius and will be performed on both the aluminum and steel. Using the data from the experiment, graphs and figures can be made comparing these two materials.

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Equipment and Procedure

Equipment for this Experiment included a Riehle PI-2 Impact testing machine, 1018 Cold Finished Steel, 6061-T6 Aluminum, liquid nitrogen, a hot pot, and an IR gun. The setup consisted of using liquid nitrogen to cool down samples and a hot plate to boil water to heat up samples. Heating the sample was easy because it used boiling water which has a boiling point of about 100 degrees which is conveniently the required temperature for my sample in this specific experiment. The cooling process involved liquid nitrogen to rapidly cool the sample and alcohol to bring the sample back up to the target temperature. Because of the strenuous nature of this process multiple samples were added. The next task was to set up the Riehle machine, bringing the Pendulum to a height of 48in and selecting the 15lb hammer. To start the test, the cooled or heated sample was loaded into the selected hammer and the pendulum was activated. At times there were moments when the sample flew back towards the operator and incredibly sparse numbers were recorded. This action would cause the experiment to be considered null, so it is important to catch the hammer on the return swing, followed by engaging the brake and then reading the data. The expended energy is read directly off the scale, located on the machine, by counting the tick marks on a dial and reading based on the selected scale. Next the flying distance was read by measuring the distance between the original point to the point where the first piece of the sample touched the ground. This involves using a Bosch DLR130k laser distance measurer and putting a book behind the sample so that the laser measures the sample distance not the wall. Do this for both the aluminum and steel measuring both at -123 degrees and 100 degrees Celsius.

Result and Discussion

6061 Aluminum Observations

In this experiment, the impact energy is measured using a Riehle PI-2 Impact testing machine which was set up as previously described. The samples of aluminum break completely from testing. In addition the center of each sample looks dull. The fact that aluminum is relatively ductile explains this effect. It is also possible that some plastic deformation is happening before the samples break. However, after observing the outside edge of each fracture we can see the trademark shine of aluminum which suggests the presence of an initial clean cut. The aluminum sample shows that the main driver of its failure is bending stress rather than shear stress. This is normal among such material because the bending stress is twice as intense as the shear stress. Aluminum does not absorb significant energy which shows that this material has a low toughness relative to that of steel. The main attribute observed regarding the aluminum sample is the consistency of its numbers despite the changing temperature. This can only lead to the conclusion that aluminum is not sensitive to temperature and will exhibit the same properties no matter the temperature.

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1018-Cold Finished Steel Observations

The observation of steel indicates that it is tougher than aluminum but much more spontaneous. Under ideal conditions, steel has a toughness double that of aluminum but only within this ideal condition. Furthermore, depending on the conditions of the test, steel can and does exhibit both ductile and brittle behaviors. Many of the samples did not break completely, only partially breaking during the test at temperatures under zero degrees Celsius. It was discovered that temperature is a condition that affects the properties of steel. When cold, steel is shown to be brittle and has a low energy absorption in addition to samples fully breaking. On the contrary, above freezing, steel shows ductile behaviors with high energy absorption and only breaks partially during the test. Steel has a transition temperature of zero degrees Celsius in which the properties will suddenly change. While steel is significantly tougher than aluminum and can absorb more energy, it can also be said that steel is extremely sensitive to temperature and is not suitable for certain environments.

The Numbers Analysis and Possible Cause

During the Impact test for both Aluminum and steel data on the impact energy, distances, and temperature are tracked in addition to being calculated. From this data a graph that graphs each material impact toughness and temperature can be made. Figure 1. shows the data comparing both Aluminum and steel on the same graph.

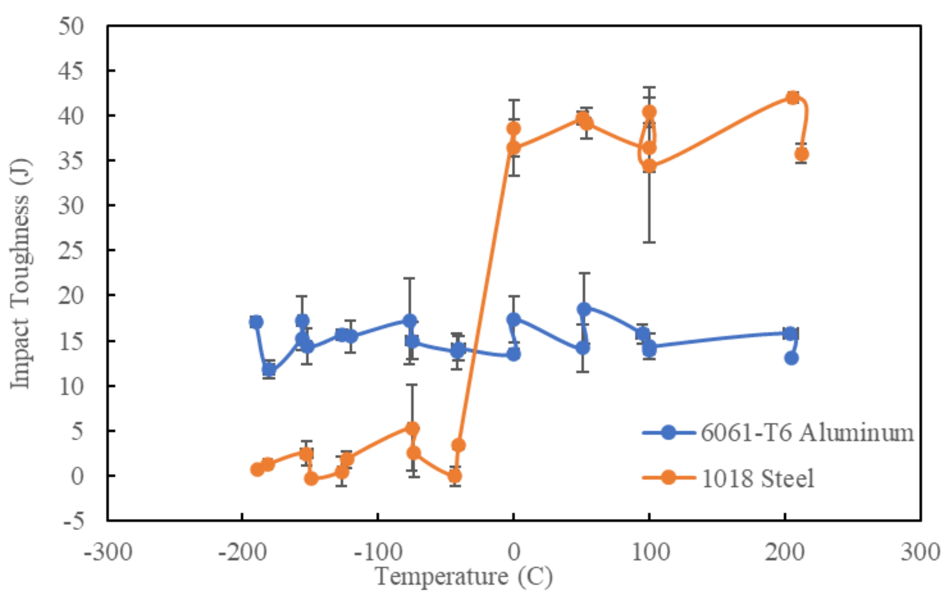

Figure 1. Impact toughness graphed against temperature of 1018-steel and 6061-T6 Aluminum

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As can be seen on the graph, Aluminum has a flat curve while the steel has an extremely aggressive curve. This shows the sensitivity of 1018 steel versus Aluminum. A cause for this difference could be the crystalline structure of each material. According to Stanislaw Zajac, PHD in Metallurgy, Aluminum has a FCC structure which negates any big differences in its properties and does not show an transition temperature behavior like a BCC structure (Zajac 2020)1 .1018 1018 Steel is a material that has a BCC structure which explains why it is so sensitive to temperature compared to the Aluminum which has a FCC structure.

Conclusions

The purpose of this experiment was to evaluate the effect of temperature on the mechanical properties of 1018 Cold Finished Steel and 6061 Aluminum. This was achieved by using a Riehle PI-2 impact testing machine on samples which were cooled or heated to the target temperature. The experiment concludes that on average steel is much tougher than aluminum. However, it was discovered that steel is also incredibly sensitive to temperature compared to aluminum whose mechanical properties are not significantly affected by temperature. The sensitivity of steel is evident in the brittle to ductile transition that occurred over differentiating temperatures. In fact, many of the steel samples only partially broke, further emphasizing the effect of temperature. In contrast, aluminum does not exhibit any ductile to brittle behavior and had the same source of failure among all samples. This could be explained by the difference in the crystalline structure of steel and aluminum; 1018 steel has a BCC structure and aluminum has a FCC structure. FCC structure does not show a transition temperature and appears to be thermally stable. Aluminum in general is shown to not to be affected by temperature and does not show a brittle to ductile transition such as steel. These findings would be especially important in material selection in many industries. Airplanes, or example, experience hot and cold temperature changes very frequently and knowing that the properties of aluminum will stay stable no matter the temperature is important to the safety of the aircraft. Conversely, knowing the property changes in steel can also yield many advantages because the steel can be modified to fit a wider range of applications using the idea of ductile to brittle transition. In conclusion, this experiment does not prove one material is better but shows the importance of the environment when selecting material.

References

1. Zajac, Stanislaw. 2020. *In the cold, aluminium gets stronger.* December 1. Accessed February 7, 2024. https://www.shapesbyhydro.com/en/expert-thoughts/in-the-cold-aluminium-gets-stronger/#:~:text=These%20changes%20occur%20in%20strength,FCC)%20crystal%20structure%20of%20aluminium.
2. The UNC Charlotte William States Lee College of Engineering Department of Mechanical Engineering and Engineering Science. 2024. "Lab Instruction_1_Impact Test of Metals_Spring 2024". Last modified Fall 2024. https://uncc.instructure.com/courses/218553/pages/topic-1-impact-test-of-metals

Appendices

Raw Data Comparing Temperature and Aluminum

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AL			
Ave. Impact Toughness	StDv. IP	Ave Temp	StDv.Temp
17.09194659	0.590677	-190.33333	0.577350269
11.78980811	0.95792	-180.5	3.535533906
15.27802121	1.330984	-156.5	0.707106781
17.26443753	2.727332	-156.33333	2.886751346
14.34526483	1.960931	-152.66667	2.886751346
15.64734454	0.587027	-126.66667	3.511884584
15.46694609	1.759062	-120.36667	1.582192572
17.14626572	4.795059	-77	1
14.9768419	2.065099	-75	4.242640687
13.80212066	1.970342	-42	0
14.15099726	1.31356	-41	4.582575695
13.5403624	0.480695	0	0
17.33179784	2.592299	0	0
14.16816105	2.591381	50.833333	0.288675135
18.54205332	3.880152	52	1
15.7538269	1.079189	94.666667	4.041451884
13.87370443	0.334113	100	0
14.32789327	1.434204	100	0
15.77550994	0.453223	204	5.291502622
13.13949643	0.290887	205	0

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Steel			
Ave. Impact Toughness	StDv. IP	Ave Temp	StDv.Temp
0.62694746	0.051484	-188.667	1.15470054
1.24847762	0.188399	-181.333	2.88675135
2.46998092	1.300991	-153	4.24264069
-0.23501615	0	-149	0
0.46359931	1.643742	-127	2.64575131
1.78796438	0.896271	-123.333	2.51661148
5.30863001	4.748752	-75.25	2.36290781
2.57524422	2.686258	-73.6667	1.52752523
-0.03881259	1.063321	-43.3333	2.88675135
3.33618454	0.323727	-41	1.73205081
38.5989733	3.174151	0	0
36.50922	3.129202	0	0
39.7053133	0.745203	50.66667	1.52752523
39.1521433	1.673113	53.66667	2.30940108
36.4477567	2.761749	100	0
40.38141	1.607475	100	0
34.48093	8.607281	100	0
42.04092	0.55317	206	1
35.77166	1.043067	212	2.82842712